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Dr. Vishwanath Karad

MIT WORLD PEACE UNIVERSITY | PUNE

TECHNOLOGY, RESEARCH, SOCIAL INNOVATION & PARTNERSHIPS

SCHOOL OF ENGINEERING AND TECHNOLOGY BULLETIN

JUL-SEP,2025

MIT-WPU,Kothrud,Pune **mitwpu.edu.in**

WITH THE BLESSINGS OF REVERED
PROF. DR.VISHWANATH D. KARAD

SHRI DR. RAHUL V.KARAD
TRUSTEE AND EXECUTIVE PRESIDENT,MAEER'S MIT GROUP OF INSTITUTIONSDR

UNDER THE GUIDANCE OF

DR. SIDDHARTH CHAKARBARTI
Dean SOET
MIT WORLD PEACE UNIVERSITY

DR. PD KHANDEKAR
CHIEF ACADEMIC OFFICER
MIT WORLD PEACE UNIVERSITY

DR. RM CHITNIS
VICE CHANCELLOR
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WORLD PEACE PRAYER

ॐ नमोजी जाद्या। वेद प्रतिपाद्या। जयजय स्वयंवेद्या। आत्मरुपा ।।१।।
देवा तुंचि गणेशु । सकलार्थ मतिप्रकाशु । म्हणे निवृत्तिदासु ।
अवधारितो जी ।।२।।

गुरुर्ब्रह्मा गुरुर्विष्णुः गुरुर्देवो महेश्वरः । गुरुः साक्षात् परब्रह्म तस्मै श्री
गुरवेनमः ।।

ॐ पूर्णमदः पूर्णमिदं पूर्णात् पूर्णमुदच्यते। पूर्णस्य पूर्णमादाय
पूर्णमेवावशिष्ते
।। ॐॐॐॐ शान्तिः । शान्तिः । शान्तिः ।।

हरि ॐ ईशा वास्यमिदं सर्वम्। यत्किंच जगत्यां जगत् ।। तेन त्यक्तेन
मुंजीथाः । मा गृधः कस्यस्विद् धनम् ।।

ॐ भूर्भुवः स्वः । तत्सवितुर्वरण्यं ।। भर्गोदेवस्य धीमहि। धियो यो नः
प्रचोदयात् ।।

सर्वेऽपि सुखिनः सन्तु। सर्वेसन्तु निरामयः सर्वेभद्राणि पश्यन्तु। मा
कश्चिद् दुःखमाप्नुयात्

द्यौः शान्तिः । अन्तरिक्षं शान्तिः। पृथ्वी शान्तिः। आपः शान्तिः ।
औषधयः शान्तिः। वनस्पतयः शान्तिः । विश्वेदेवाः शान्तिः । ब्रह्म
शान्तिः। सर्व शान्तिः। शान्तिरेव शान्तिः । साऽमा शान्तिरेधि ।।१७।।
ॐॐॐ शान्तिः । शान्तिः । शान्तिः ।

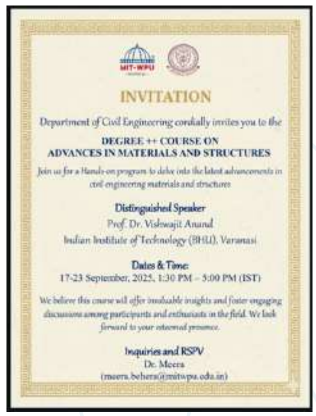
Department Initiatives & Strategic Developments

Dept. of Petroleum-

A seminar on Oil & Gas field operation was conducted by Mr Bhumin Baraiya,AEE (Drilling),ONGC organized by Dr Kumar Abhijeet Raj

Department of Civil

The Department of Civil Engineering successfully organized a one-week Degree++ Certificate Course on “Advances in Materials and Structures” in collaboration with IIT (BHU), Varanasi. The program focused on emerging construction technologies, sustainable materials, and Industry 5.0–aligned practices. It featured expert lectures and hands-on training on advanced material characterization and structural analysis. The event witnessed enthusiastic participation from postgraduate students, PhD scholars, and faculty members, fostering strong academic–industry–research integration.



Dept. of Polytechnic

The MIT WPU Department of Polytechnic, in collaboration with Aspire, successfully conducted and completed a series of multidisciplinary certificate skill courses under the National Skills Development initiative. These courses were designed to enhance practical knowledge and industry-relevant skills across different engineering disciplines. The Computer Science and Engineering (CSE) department offered a specialized course in Artificial Intelligence and Business Analytics, equipping students with advanced skills in AI technologies and data-driven decision-making. The Mechanical Engineering (Mech) department focused on Electric Vehicles (EV), providing in-depth training on the latest developments in sustainable transportation technology. Meanwhile, the Electronics and Communication Engineering (ECE) department, specifically under the Artificial Intelligence and Machine Learning (AIML) specialization, conducted a course on Embedded Systems, enabling students to gain expertise in designing and programming embedded hardware and software solutions. These multidisciplinary programs have greatly contributed to skill enhancement for the students involved.

Faculty Research Articles

Department of Civil Engineering

Flexural behavior of corroded reinforced concrete beams with GGBS: A comparative"

The flexural behavior of corroded reinforced concrete beams incorporating GGBS shows improved performance compared to conventional concrete beams. GGBS reduces concrete permeability and slows reinforcement corrosion, leading to better stiffness, higher residual flexural strength, and delayed crack initiation. Even under corrosion effects, GGBS-based beams exhibit improved bond strength, smaller crack widths, and better energy absorption. Overall, GGBS enhances durability and flexural performance while promoting sustainable construction.

Dr. Sandeep Sathe
(Department of Civil)

Dr. Ketaki H. Kulkarni)
(Department of Civil)

"Crestless contracted V-notch weir for flow measurement in triangular open channels"

The crestless contracted V-notch weir is an efficient device for measuring low flows in triangular open channels. The absence of a crest reduces energy loss and sediment accumulation, while the contracted V-notch improves sensitivity and accuracy at small discharges. Studies indicate stable discharge coefficients, lower head requirements, and reliable performance over a wide range of flow conditions. Its simple design, ease of installation, and low maintenance make it suitable for laboratory, irrigation, and environmental flow measurement applications. (

Department of polytechnic

"The AI Revolution in Web Development: 2025 and Beyond"

In 2025, Artificial Intelligence has transformed web development from a supportive tool into a powerful driving force behind innovation, efficiency, and highly personalized user experiences. AI-powered tools like GitHub Copilot and OpenAI Codex accelerate development by automating repetitive tasks, suggesting optimized code, and reducing errors, allowing developers to deliver projects faster. At the same time, machine learning enables real-time personalization, intelligent UI/UX adjustments, predictive analytics, and automated testing to improve performance and reliability.

Beyond coding and design, AI is reshaping content creation by generating relevant and engaging material based on user behavior and preferences. It also strengthens cybersecurity through real-time threat detection and enhances accessibility with voice and visual search integration. Overall, AI is redefining every stage of web development, empowering businesses to build smarter, faster, and more user-centric digital experiences.

Mrs. P.S.Patil
(Department of Polytechnic)

Mrs. M.A.Dhotay
(Department of Polytechnic)

"Adaptability: The Ultimate Power Skill for Today's Professionals"

Forget the hype—coding isn't the only skill of the future. Neither is AI. The real game-changer is adaptability: the ability to stay resilient and grow in a constantly changing world.

Today's workplace is evolving rapidly. Roles are being redefined, industries disrupted, and traditional career paths reshaped. In this environment, technical skills alone aren't enough. What truly sets professionals apart is their ability to embrace change, handle uncertainty, and keep evolving.

Adaptability means being proactive—learning, unlearning, and relearning as new trends emerge. Whether it's marketers using AI tools, teachers leading virtual classrooms, or leaders managing hybrid teams, those who thrive don't just react to change—they lead it.

At its core, resilience fuels adaptability. It helps professionals turn setbacks into growth opportunities and move forward stronger than before.

Dept. of Mechanical Journal Publications:

Faculty Name	Authors	Title	Year	Source title	Volume & Issue	Page No
Dr. S. B. Barve	Pesode P.; Barve S.	A review-recent advances on antibacterial coating on magnesium alloys by micro arc oxidation for biomedical applications	Aug 2025	Advances in Materials and Processing Technologies	Vol-11,Issue-3	1475-1517
Dr. M. M. Lele	Dhavale A.A.; Lele M.M.	Optimizing thermal performance in tube-in-tube heat exchangers with metal foam inserts: experimental and RSM analysis	Oct 2025	Experimental Heat Transfer	Vol-3325,Issue-1	-
Dr. M. M. Lele	Dhavale A.A.; Lele M.M.	Numerical investigations on the impact of metallic foam configurations on heat transfer in double tube heat exchanger: A parametric approach	Mar 2025	Numerical Heat Transfer; Part A: Applications	Vol-,Issue-	-
Dr. M. M. Lele	Dhavale A.A.; Lele M.M.	Enhancing thermal-hydraulic performance in heat exchangers with metal foam inserts: a comprehensive experimental investigation	May 2025	Experimental Heat Transfer	Vol-16,Issue-5	73-108
Dr. D. P. Hujare	Kulkarni S.; Kulkarni M.; Hujare D.	A comprehensive acoustic analysis of a single expansion chamber reactive muffler	Jun 2024	Building Acoustics	Vol-28,Issue-1	19-74

Dr. D. R. Waghole	Kongi P.; Waghole D.R.; Babu P.K.A.	Enhanced thermal management of 21,700 NMC Li-ion batteries using PCMs: correlation development and numerical analysis	Aug 2025	Ionics	Vol-38,Issue-6	592-654
Dr. P. N. Dhatrak	Chaudhari S.; Khade A.; Girase V.; Dhatrak P.	Recent Advancement and Development of Biomimetic Heart Valve Prosthesis	Aug 2025	Journal of Bionic Engineering	Vol-86,Issue-16	5682-5734
Dr. P. N. Dhatrak	Kulkarni A.; Raut R.; Dhatrak P.	A comprehensive review on configuration, design and programming of robotic systems used in various applications	Aug 2025	International Journal of Intelligent Robotics and Applications	Vol-38,Issue-6	540-591
Dr. P. N. Dhatrak	Dhatrak P.; Durge J.; Dwivedi R.K.; Pradhan H.K.; Kolke S.	Interactive design and challenges on exoskeleton performance for upper-limb rehabilitation: a comprehensive review	Feb 2025	International Journal on Interactive Design and Manufacturing	Vol-29,Issue-2	154-163
Dr. P. N. Dhatrak	Deshmukh P.; Dhatrak P.	Influence of Marginal Bone Loss with Variable Mandibular Bone Densities Under Masticatory Loading Using Finite Element Analysis	Oct 2025	Regenerative Engineering and Translational Medicine	Vol-3325,Issue-1	-
Dr. P. N. Dhatrak	Raibole K.V.; Sonawwanay P.D.	Factors affecting and methods of reducing thermal stratification in cryogenic storage tanks of launch vehicles: review article	Feb 2025	Journal of Thermal Engineering	Vol-,Issue-	73-85

Dr. P. M. Patane	Kulkarni G.S.; Siddeshkumar N.G.; Prasad C.D.; Shankar B.L.; Aprameya C.R.; Patane P.; Deshmukh U.D.; Prasad C.	Synthesis and Investigation on Mechanical Properties of Hybrid FRP Composite Using Taguchi Technique	Aug 2025	Journal of The Institution of Engineers (India): Series D	Vol-,Issue-	-
Dr. G. D. Shrigandhi	Kumar L.J.; Ganjigatti J.P.; Irfan G.; Thara R.; Aradhya P.J.; Choudhary R.K.; Shrigandhi G.D.; Gurunagendra G.R.	Effect of Graphene and Coconut Shell Ash on Mechanical Properties of Al6061 Metal Matrix Composites	Dec 2025	Journal of The Institution of Engineers (India): Series D	Vol-72,Issue-1	-
Dr. C. G. Burande	Sharma S.; Meena P.K.; Tripathi A.K.; Burande C.G.; Shelare S.; Wagle C.S.	Advances in photo-fermentative biohydrogen and biomethane production from organic waste: a comprehensive review of processes, parameters, and integrated pathways	Oct 2025	Biofuels	Vol-3325,Issue-1	-
Dr. Parshant Kumar	Kumar H.; Prasad R.; Kumar P.	Characterization of copper based composite reinforced with SiC via FSP	Oct 2025	AIP Conference Proceedings	Vol-3325,Issue-1	-
Dr. Parshant Kumar	Kumar H.; Prasad R.; Kumar P.	Enhancement of strength and ductility of aluminum-silicon alloy by FSP	Feb 2025	AIP Conference Proceedings	Vol-,Issue-	185-199

Dr. Parshant Kumar	Kumar H.; Prasad R.; Kumar P.; Deo M.	Characterization of GTAW welded steel (572 GR 50)	Jun 2025	AIP Conference Proceedings	Vol-11,Issue-1	99-114
Dr. M. T. Solanki	Patel V.; Behera A.; Solanki M.; Thomas T.; Azim S.	Mechanical Properties and Behaviour of Functionally Graded Materials	Oct 2025	Novel Applications of Functionally Graded Materials	Vol-3325,Issue-1	-
Dr. M. T. Solanki	Solanki M.T.; Vakharia D.P.	A Critical and Comparative Study of Layered Cylindrical Hollow Roller Bearing Using Numerical and Experimental Analyses for the Assessment of Static Load Carrying Capacity	Oct 2025	Journal of Tribology	Vol-3325,Issue-1	-
Dr. Vivek Patel	Behera A.; Patel V.; Thomas T.; Makireddi D.; Dehury J.; Mohapatra S.R.	Fatigue Behavior of Functionally Graded Honeycomb Sandwich Composites	Jun 2025	Novel Applications of Functionally Graded Materials	Vol-175,Issue-	-
Dr. B. Sahoo	Jain R.; Sahoo B.; Jain S.; Mohan M.; Choudhary M.; Lee H.; Dewangan S.K.; Samal S.; Ahn B.; Jeon Y.	Functionally Graded Metallic Materials Via Additive Manufacturing: Research Progress on Processing, Challenges, and Applications	Jan 2025	International Journal of Precision Engineering and Manufacturing - Green Technology	Vol-,Issue-	149-179

Department of EEE

Authors	Title	Year	Source Title	Volume & Issue
Ashish Joshi, Kaushik Joshi, Rupali Kute, Sunil Jaiswal, Prerna Mishra	Unleashing the Power of Deep Learning: A Novel Approach for Defect Detection in Solid Rocket Motor	2025	FirePhysChem	–
B. Paranjape, A. Naik, S. P. Sankar	A Performance Comparison of Object Detection Algorithms on Traffic Scenes in Indian Roads	2025	Engineering, Technology & Applied Science Research	Vol. 15, Issue 4

Department of Civil Engineering

Authors	Title	Year	Source Title	Volume & Issue	Page No / Article No
S. Sathe et al.	Flexural behavior of corroded reinforced concrete beams with GGBS: A comparative study	2025	Iranian Journal of Science and Technology, Transactions of Civil Engineering	–	–
H. Kadam, J. Patil, P. Minde	Comparative analysis of advanced bracing systems for lateral stability of RCC bridge bents under seismic excitation	2025	Journal of Building Pathology and Rehabilitation	Vol. 10, Issue 2	–
U. J. Patil et al.	Influence of infill wall opening on seismic performance of irregular RC buildings using adaptive pushover analysis	2025	Journal of Building Pathology and Rehabilitation	Vol. 11, Issue 1	–

A. D. Shitole et al.	Investigation of mechanical performance of concrete with end-of-life solar panels as sand replacement	2025	Journal of Sustainable Cement-Based Materials	—	—
M. P. Shaikh et al.	Integration of frequency analysis and HEC-RAS for probabilistic flood hazard mapping	2025	Proceedings of the Institution of Civil Engineers: Water Management	—	—
L. Gautam et al.	Environmental impact mitigation and durability enhancement of concrete through fly ash substitution: A comprehensive review	2025	Journal of Building Pathology and Rehabilitation	Vol. 10, Issue 2	—
N. S. Pawar, K. V. Sharma	Advanced fuzzy-based landslide prediction in high-risk regions using hybrid and statistical GIS models	2025	Physical Geography	Vol. 46, Issue 4	307–337
P. Minde, J. Patil	Navigating the path to construction's digital renaissance	2025	Digital Transformation in the Construction Industry (Elsevier)	Book Chapter	45–59
B. Achour, L. Amara, K. H. Kulkarni	Crestless contracted V-notch weir for flow measurement in triangular open channels	2025	Larhyss Journal	—	Article No. 11123680

Department of Petroleum

Authors	Title	Year	Source Title
Jyotsna; G. D. D. Singh; A. Kushwaha; S. Kumar; R. Kumar; L. Puppala; B. G. Desai	Characterization of dual porosity in bioturbated tight sandstone reservoir: Insight from Mesozoic Jhuran Formation of Kachchh–Saurashtra offshore basin, India	2025	Acta Geophysica

Department of Polytechnic

Authors	Title	Year	Source Title	Volume & Issue	Page No
R. Kale; A. S. Gaadhe; N. B. Pokale; P. A. Patil	Design and implementation of secure cryptographic algorithms for wireless sensor networks in healthcare applications	—	—	—	—
Y. Gaikwad; S. Bhattacharya; M. Gulhane; A. Sharma; N. Rakesh; D. Dhabliya	Improving Healthcare with Remote Patient Monitoring and Real-Time Data Analytics Using IoT	2024	Proceedings of the 6th International Conference on Information Management & Machine Intelligence	Conference	1–9
S. Malwade; R. Jadhav; V. Jadhav; A. V. Chitre	Blockchain-based secure data sharing and storage system using elliptic curve cryptography	—	—	—	—
S. Budhavale; L. R. Desai; S. A. Paithane; S. N. Ajani	Elliptic curve-based cryptography solutions for strengthening network security in IoT environments	—	—	—	—
H. S. Ohal; S. Mantri	A Comprehensive Analysis of Machine Learning Algorithms for Predictive Modeling in Dementia Detection	2025	Journal of The Institution of Engineers (India): Series B	Vol. 106, Issue 5	1673–1689
S. H. Kulkarni; V. V. Deshmukh; R. K. Moje; S. H. Lavate	Applied discrete mathematics in developing efficient cryptographic algorithms for enhancing information security in WSN	—	—	—	—

Department of Chemical Engineering

Authors	Title	Year	Source Title	Volume & Issue	Page No / Article No
G. Kalaiyan; N. S. Topare; S. Sikiru; S. Thambidurai; M. Kandasamy; N. Pugazhenthiran; C. Shanthi; P. Rameshkumar; S. Suresh	Green synthesis of copper oxide nanoparticles using *Euphorbia heterophylla* leaf extract for deactivation of pathogenic bacteria and photocatalytic degradation of industrial dyes	2025	Results in Engineering	Vol. 26	Article No. 104797

Book chapter

Department of EEE

Authors	Title	Year	Source Title	Volume & Issue
Netra Lokhande; Sharvari Dhamale; Chinmayi Railkar	Chapter 5 – Flex sensors in healthcare communication: Accessibility and interaction	2025	*Rehabilitation Engineering, Assistive Technologies and the Future of Healthcare Using Robotics* (Academic Press)	Book Chapter

Department of Polytechnic

Authors	Title	Year	Source Title	Volume & Issue	Page No
J. Mante; K. Kolhe	Introduction to IoT and Security Issues	2025	*Fostering Machine Learning and IoT for Blockchain Technology: Smart Cities Applications* (Springer Nature Singapore)	Vol. 1 (Book Chapter)	283–326
N. Dongre; J. Mante; M. Aware; P. Nehete; M. Dhotay; K. Chavhan	Smart emotion based music player system using face detection	—	*Recent Advances in Sciences, Engineering, Information Technology & Management* (CRC Press)	Book Chapter	765–769
S. S. Palimkar; S. S. Malwade; F. I. Shaikh; B. Nagalakshmi; L. J. Grace; K. S. Sidhu	Integrating Machine Learning With Secure Telemedicine Image Transmission	2025	*Advanced Secure Transmission of Telemedicine-Based Bio-Medical Images* (IGI Global Scientific Publishing)	Book Chapter	115–138

Department of Mechanical

Author	Title of the Book	Name of Publisher & place	Date of Publication
Patel V.; Behera A.; Solanki M.; Thomas T.; Azim S.	Mechanical Properties and Behaviour of Functionally Graded Materials	CRC Press	Sep 2025
Behera A.; Patel V.; Thomas T.; Makireddi D.; Dehury J.; Mohapatra S.R.	Fatigue Behavior of Functionally Graded Honeycomb Sandwich Composites	CRC Press	Sep 2025

Patent/ IPR

Department of EEE

Faculty Name / Inventor	Patent Title	Application No.	Year	Status	Publishing Authority
Dr. V. N. Deshmukh; Dr. Subramaniam Radhakrishnan	System for thermal management of battery for vehicle	2021211029238	2025	Published on 25 September 2025	Government of India

Department of Mechanical

Patent Application No.	Inventors Name	Title of the Patent	Patent Published / Granted Date
202521057027	CHINMAY KANE, PUSKARAJ D SONAWWANAY	A BAR MAKING DEVICE	7/4/2025
202521057482	RAHUL ASHOK PATIL, DR VAIBHAV NARENDRA DESHMUKH	A SOLAR-POWERED EVAPORATION DEVICE	7/4/2025
202521058576	Dr. Deepak Popat Hujare, Dr Girish Sharad Barpande, Dr Mahesh V. Kulkarni	A SYSTEM AND METHOD FOR PRECISE SIMULATION OF UNBALANCE AND MISALIGNMENT FAULTS IN ROTATING MACHINERY	7/11/2025
202521060837	Dr. KUNDLIK VITTALRAO MALI, Mr. ASHOK BALKRISHNA KORANE	A SUCTION-SIDE REFRIGERATED AIRDRYING WITH PCM STORAGE	7/18/2025
202521060779	Sagar Govind Mushan, Dr. Vaibhav Narendra Deshmukh	A PERFORMANCE EVALUATING APPARATUS FOR HEAT PIPES	7/11/2025
202424065764	BONDAR SHRIHARI MAHADEO, MAHESHWARI KUNJ, MALVIYA AYUSH, PAREKH VYOM	A BRAKE SYSTEM SAFETY DEVICE FOR ELECTRIC VEHICLES	7/11/2025

202521064691	Jay Kirdat, Rahul Jagtap	INTERNAL SCREW THREAD INSPECTION MACHINE	7/25/2025
202521064685	DR. SANDEEP TUKARAM CHAVAN, AMARJEET YADAV, DNYANSHRI CHAUDHARI, SHRIKANT NIMBALKAR, SNEHAL KOLEKAR	A SYSTEM FOR CONTROLLING TRAFFIC JAMMER	7/25/2025
202521066957	MRS. PALLAVI SANTOSH MITKARI, DR. ABHISHEK MADHUKAR THOTE, DR. THIRUGNANAM ARUNACHALAM, DR. VIKRANT JADHAV	A DEVICE FOR PALATAL EXPANSION	8/1/2025

Department of Polytechnic

Sr. No.	Patent Title	Inventor(s)	Application No. / Ref. No.	Status
1	E-Learning Recommendation System based on Feedback	Prof. Dr. R. S. Kale	202121000385	**Granted**
2	Stick for elderly people	Prof. Vaibhav Rangari; Prof. Mangesh Mahajan	202521079185	**Filed**
3	A system and method for greenhouse monitoring	Prof. Naresh B. Chaudhari; Prof. Sachin N. Patil	202521082813	**Filed**
4	Interlocking Hollow paving block mould *(Design Patent)*	Prof. Sagar Sonawane	Ref. No. 463604-001	**Filed**

Sr. No.	Copyright Title	Author / Inventor	ROC No.	Status
1	Diagrammatic representation of component parts of compaction factor test equipment	Prof. Sagar Sonawane	6555/2024-CO/A	**Registered**

Department of Chemical Engineering

Sr. No.	Design Title	Inventor(s)	Registration No.	Status	Date
1	Photocatalytic Reactor for Effective Wastewater Treatment	N. S. Topare; P. Polshettiwar; P. Chavan; N. Arshid; A. Khan; K. Muthusamy; P. Gunnasegaran; B. Kale	470379-001	**Design Registration Granted (India)**	20-08-2025

Department of Bio-Engineering

Sr. No.	Design Title	Inventor(s)	Registration No.	Status
1	A germicidal compostio for vegetable & fruit wash using hydrophobin	Dr. Shraddha Kulkarni	549852	**Granted**

Indian Design Registration Granted

N S. Topare, P Polshettiwar, P Chavan, N Arshid, A Khan, K Muthusamy, P Gunnasegaran, B Kale, Photocatalytic Reactor for Effective Wastewater Treatment, Indian Design Registration Granted 470379-001 (20/08/2025).

P Polshettiwar
M.Tech Student

Prof. N S. Topare
Assistant Professor

N S. Topare, P Polshettiwar, P Chavan, N Arshid, A Khan, K Muthusamy, P Gunnasegaran, B Kale, Photocatalytic Reactor for Effective Wastewater Treatment, Indian Design Registration Granted 470379-001 (20/08/2025).

Indian Design Registration Granted

N S. Topare, A Kale, P Chavan, C Sayankar, N Arshid, S A Khan, K Muthusamy, B Kale, A Batch Reactor System for Fast and Efficient Conversion of Triglycerides into Biodiesel, Indian Design Registration Granted 470251-001 (19/08/2025).

A Kale
M.Tech Student

Prof. N S. Topare
Assistant Professor

N S. Topare, A Kale, P Chavan, C Sayankar, N Arshid, S A Khan, K Muthusamy, B Kale, A Batch Reactor System for Fast and Efficient Conversion of Triglycerides into Biodiesel, Indian Design Registration Granted 470251-001 (19/08/2025).

Indian Design Registration Granted

V H. Raghatate, P T. Mali, N S. Topare, C Sayankar, S Landge, S Naikwadi, System With Unique Internals for Oil and Gas Separations, Indian Design Registration Granted 457498-001 (28/08/2025).

V H. Raghatate
B.Tech Student

P T. Mali
B.Tech Student

Prof. N S. Topare
Assistant Professor

V H. Raghatate, P T. Mali, N S. Topare, C Sayankar, S Landge, S Naikwadi, System With Unique Internals for Oil and Gas Separations, Indian Design Registration Granted 457498-001 (28/08/2025).

Faculty Achievements & Recognition

Department of EEE

Dr. Apurva Naik, Associate Professor in Department of Electrical and Electronics Engineering received the Ideal Teacher Foundation Day award 2025 on 5th September 2025 by the MAEER's Group.2025 held on 19, 20 July 2

Department of Mechanical

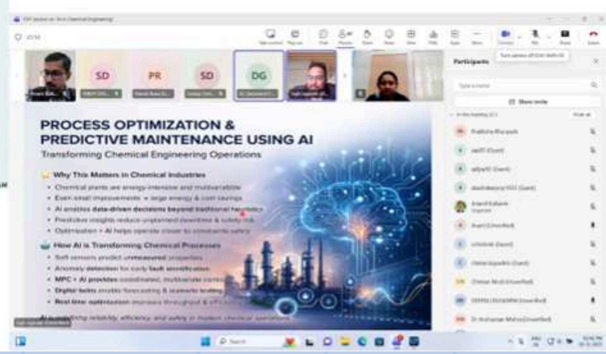
Dr. Omkar Kulkarni, IEI NMLC–FCRIT Excellence Award 2025. The Institution of Engineers (India), Navi Mumbai Local Centre and Fr. C. Rodrigues Institute of Technology. Teaching Excellence Award 2025

Department of Chemical Engineering

Prof. Niraj S. Topare initiated the Student Exchange Agreement between MIT World Peace University and University of Regina, Canada

Inauguration of the Chemical and Biological Analysis Lab by **Prof. Jeff Keshen**, President and Vice-Chancellor

5 Day FDP on AI in Chemical Engineering: Industry, Curriculum and Research Perspectives



Department of Civil

Dr. Pravin Minde, Assistant Professor in Civil Engineering at MIT World Peace University, Pune, has demonstrated significant academic, research, and professional achievements. He actively contributed to international academia by guiding postgraduate students, notably supporting international conference presentations such as at the 3rd International Conference on Sustainable Buildings and Construction (ICSBIC 2025) in Muscat, Oman, highlighting emerging technologies like drone applications in construction.Dr. Minde strengthened his professional expertise by completing a 7-day Faculty Development Programme on Construction Law, Contracts, and Arbitration, enhancing his domain knowledge in construction management and legal compliance. He also served as a reviewer for reputed international conferences and was honored with membership on the Editorial Board of Scientific Reports (Springer Nature), a prestigious Q1, SCIE-indexed journal.

Energy Efficiency (2025) 18:98
https://doi.org/10.1007/s12053-025-10392-4

RESEARCH

Life cycle energy assessment of conventional and modular buildings: a case study analysis in the Indian context

Pravin Minde · Kaushik Kore

Received: 15 November 2024 / Accepted: 22 October 2025

Abstract Precast construction provides better quality control, improved site safety, and faster project timelines than traditional RCC (Reinforced Cement Concrete) methods. The push for sustainable building practices has increased in India, yet few studies offer a detailed comparison between conventional RCC buildings and modular alternatives. This study addresses that gap by conducting a comprehensive life cycle energy assessment (LCEA) of both construction types (Traditional RCC building and Light gauge steel frame (LGSF)-Ferron Modular Building in a mid-rise residential project in Maharashtra. Using a hybrid approach that integrates process-based and input-output methods, the study evaluates embodied, operational, and demolition energy, offering valuable insights into the energy efficiency of modular construction in the Indian context. Findings show that operational energy accounts for ~83% of total life cycle energy, followed by embodied (~15%) and demolition (~2%). The LGSF-Ferron building consumes 11.06% less life cycle energy (18,486.54 MJ/m^2) than RCC (20,787.26 MJ/m^2), mainly due to reduced embodied energy. This difference is specific to the case study and may vary with regional factors. The study provides evidence-based insights into material choices, construction methods, and energy-efficient strategies that contribute to India's sustainable development goals. However, broader sustainability claims require further analysis of additional environmental impacts. The study also offers actionable insights for architects, engineers, and policymakers to explore energy-efficient modular construction, contributing to sustainable development in India.

Keywords Precast construction · Life cycle energy assessment · Embodied energy · RCC (Reinforced Cement Concrete) · LGSF (Light Gauge Steel Frame) · Energy-efficient structures

Introduction

Construction projects often involve long periods, complex resources, many participants, and volatile variables. Energy is continuously consumed throughout the construction industry's supply chain and at every stage of a building's life cycle (Dahiya & Laishram, 2024). Over the past 200 years, massive population growth has hurt the environment and the planet's

His research excellence is reflected through the publication of articles in SCIE, Q2-ranked Springer Nature journals and conference proceedings published by Taylor & Francis. Additionally, Dr. Minde played a key organizational role as Convenor and organizing team member for international conclaves and workshops at MIT-WPU, fostering collaboration among academia, industry, and policymakers.Through training programmes for BRO officers, expert lectures, alumni connect sessions, and industry visits, Dr. Minde consistently bridged theory and practice, contributing meaningfully to research, education, and nation-building initiatives.

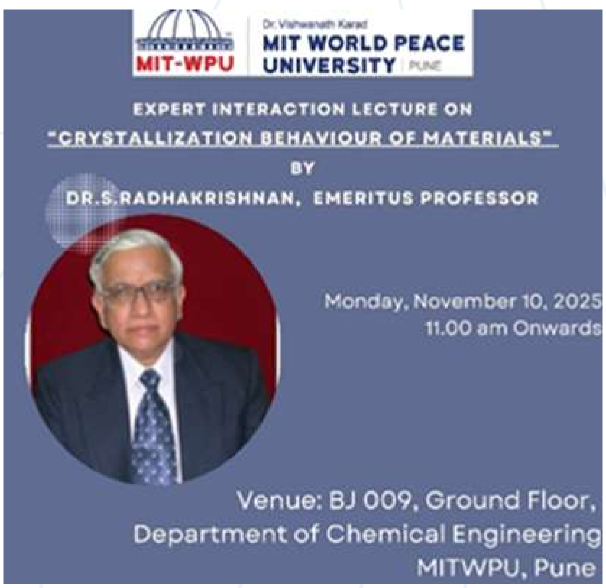
Industry Collaborations

Department of EEE

a) Department of Electrical and Electronics Engineering has signed the MoU with VLSI society of India to expose students with real world VLSI design and semiconductor technologies, research opportunities and form VSI chapter at the University.

b) Department of Electrical and Electronics Engineering has signed the MoU with Vivasvat Revolutions Limited for technology transfer and training programs in blockchain technology for financial domain.

Department of Chemical Engineering



Dr. Minde, Dr. Jagruti, Dr. Shelke, Dr. Yadav had conducted a training program for BRO officers.

This is to certify that the faculty member served as Session Chair at the International Conference on Civil, Environmental and Applied Sciences (ICCEAS-2025).

CERTIFICATE
OF SESSION CHAIR

This is to certify that to

Dr. Lipsa Mishra (Dr. Vishwanath Karad MIT World Peace University, Pune)

has chaired a technical session at the **International Conference on Civil, Environmental and Applied Sciences** (**ICCEAS-2025**), organised by the Department of Civil Engineering and Basic Science & Humanities, Gandhi Institute for Education and Technology, Baniatangi, during 18-19 July, 2025.

Dr. S Sankalp	Dr. S R Mishra	Dr. B K Panigrahi	Dr. J Jena	Dr. J Mishra
Convenor	Convenor	Conference Chair	General Chair	Patron

Publishing Partner: Cureus, Scopus

In Academic Partnership with: SRM, CHRIST, SRKR

Department of Polytechnic

We are delighted to announce that Prof. Jyoti Mante (Khurpade), Programme Head CSE & AI-DS , Department of Polytechnic, has been featured among the Top 50 Women Power Leaders by Citiesabc,Businessabc-Global AI Business Directory, Ztudium, London. This recognition highlights her remarkable contributions as an AI-cybersecurity researcher, Advisory Board Member at PHN Technology Pvt. Ltd., and her excellence as an award-winning academic and educator. Her unwavering commitment to innovation, leadership, and empowerment in the field of technology continues to inspire both students and colleagues.


MIT-WPU
MIT WORLD PEACE UNIVERSITY | PUNE
Department of Chemical and Materials Science & Engineering
We are honored to welcome Dr. Natu Varun, Scientist at CSIR-National Chemical Laboratory (NCL), for a guest lecture on Mxene and its synthesis techniques. Dr. Varun earned his PhD in Material Science and Engineering from Drexel University and is a highly accomplished researcher, having been recognized among the World's Top 2% Scientists.
Guest Lecture On :
Topic- Mxene and its synthesis techniques
Date- 10th November
Time- 3:00 PM
Venue- BJ009

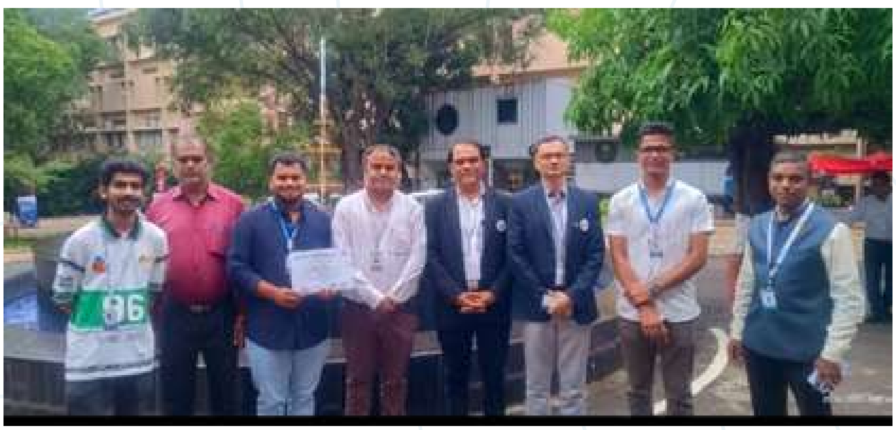

MIT-WPU
CELEBRATING NATIONAL METALLURGIST'S DAY
14TH NOVEMBER 2025
"From ore to alloy, from idea to innovation"
JOIN THE EXPERT SESSION BY METACRYST
SHRI SANJAY KULKARNI
PLANT METALLURGIST, AMERICAN AXLES AND MANUFACTURING CORPORATION LTD
EXPERTISE:
• HEAT TREATMENT PROCESS OPERATIONS & QUALITY CONTROL
• METALLURGICAL TESTING AND EVALUATION OF METALS
• FORGING AND CASTING
• CASE HARDENING INSPECTION
• FAILURE ANALYSIS AND PREVENTIVE ACTIONS
TOPIC:
RECENT DEVELOPMENTS IN HEAT TREATMENT AND FUTURE JOB PROSPECTS IN HEAT TREATMENT INDUSTRY FOR MATERIALS ENGINEERS.
TIME: 2:00 PM - 3:30 PM
VENUE: BJ 309


MIT-WPU
Dr. Vishwanath Karad MIT WORLD PEACE UNIVERSITY | PUNE
Department of Chemical and Materials Science & Engineering
Guest Lecture
on
"Tetrapods-based Smart Materials for Advanced Technologies"
2.30 - 4.30 pm on 23rd Sept 2025
Prof. Yogendra Kumar Mishra, Ph.D. Dr. habil., FRSC
Smart Materials, Mads Clausen Institute, University of Southern Denmark
Link to join
meet.google.com/rox-nzsm-egz
Attend Positively


MIT-WPU
MIT WORLD PEACE UNIVERSITY | PUNE
Department of Chemical and Materials Science & Engineering
EXPERT LECTURE ON-
"An Introduction to the Semiconductor Industry, Manufacturing Process, and the Role of Materials Engineers"
BY
Mr. Mukesh Jain
Semiconductor Engineer, Micron Technology, Taiwan
B.E. Polymer Engg. MIT Pune (2020 Batch)
M.Tech Materials Engg. IISc Bangalore
Saturday, November 15, 2025
11:00 AM (IST)
Mode: Online
Join us for an insightful session where Mr. Mukesh Jain will share an overview of the semiconductor industry, discuss the manufacturing process, his role as a Contamination Control Engineer, and opportunities for materials engineers in this field.

Department of Bio-engineering

Team ‘Udajana’ presented and won 1st Prize in Poster presentation on Production of Green Hydrogen at 2nd National Scientists Round Table Conference NSRTC 2025 under guidance of Dr. Sagar Kanekar and Dr. Anand Kulkarni.



MIT-WPU MIT WORLD PEACE UNIVERSITY | PUNE

NATIONAL SCIENTISTS ROUND TABLE CONFERENCE (NSRTC 2025) FOR VIKSIT BHARAT 2047

NSRTC

Certificate of Merit

This certificate is proudly presented to

Nirav Elebariya and team

for Poster Presentation, securing 1st / 2nd / 3rd position at

NATIONAL SCIENTISTS ROUND TABLE CONFERENCE FOR VIKSIT BHARAT 2047

from 18th to 20th July 2025

at Dr. Vishwanath Karad MIT World Peace University, Pune, Bharat.

Your gracious presence is appreciated as we celebrate the groundbreaking research and contributions of our esteemed scientists.

It is a matter of great pride and pleasure to Congratulate you in the Month of July 2025, on behalf of Hon'ble Dr. Rahul V. Karad, Executive President, Board of Directors, and Trustees of Dr. Vishwanath Karad, MIT World Peace University, Pune, Bharat.

Best Always,

Prof. Dr. Anup Kale	Prof. Dr. Siddharth Chakrabarti	Prof. Dr. Bharat B. Kale, FRSC	Prof. Dr. Murthy Chavali
Co-Convener, NSRTC 2025	Co-Convener, NSRTC 2025	Convener, NSRTC 2025	Convener, NSRTC 2025
Dean In-charge	Dean		
School of Science and Environmental Studies	School of Engineering and Technology		
MIT-WPU	MIT-WPU	MIT-WPU, Pune	MIT-WPU

Initiated Industry Partnership & MOU

Sr. No.	Company Name	Contact person name and details	MIT WPU SPOC name and contact details	MoU signing date	MoU validity details
1	S-Cube Mass Transfer, Pune	Mr. Chetan Sayankar (Founder & CEO)	Prof. Niraj S. Topare (SPOC & MoU Coordinator)	24/11/2024	24/11/2029 [05 year]
2	University of Regina, Canada	Mr. Haroon Chaudhry (Ass. Vice President)	Prof. Niraj S. Topare (SPOC & MoU Coordinator)	03/02/2025	03/02/2030 [05 year]
3	Inverted Energy Pvt, Ltd, Delhi	Mr. Rahul Raj (Director)	Prof. A D. Kulkarni (SPOC)	13/09/2024	13/09/2029 [05 year]

Industrial Visit

Department Of EEE

The Electrical and Computer Engineering students along with Dr. Netra Lokhande and Prof. Shantaram Dond of Department of Electrical and Electronics of MIT WPU visited on 7th August 2025. Nissar Transformers Private Limited in Shirwal, Pune. Nissar Transformer has extensive specialization in manufacturing distribution and power transformers with MSEB as a major customer.

Department of Civil Engineering

1. The National Academic Immersion Program (NAIP) 2024–25 at Konkan Railway Academy provided TY B. Tech Civil Engineering students with practical exposure to real-time railway infrastructure and operations. The program included expert sessions and site visits on track engineering, signaling systems, bridges, tunnels, maintenance depots, and logistics. Students interacted with KRCL engineers to understand engineering challenges in complex terrains. The immersion effectively bridged classroom theory with field practices. Overall, the program enhanced technical knowledge, safety awareness, and professional readiness.

2. An industrial site visit was conducted to Structus Consultants Pvt. Ltd., Pune, as part of the postgraduate course on Design of High-Rise Structures. The visit provided practical exposure to the design philosophy, construction methodology, and execution of high-rise RC buildings. Students observed reinforcement detailing, use of mechanical couplers, hidden beams, and advanced QA/QC practices. The visit also highlighted the importance of safety management, geotechnical integration, and BIM-based coordination. Overall, it effectively bridged theoretical concepts with real-time structural engineering practices.

3. The industrial visit to the CLOUD 28 construction site in Bhugaon, Pune focused on observing RCC structural elements such as shear walls, retaining walls, columns, beams, slabs, and an underground water tank. Students connected theoretical advanced structural design concepts with real-world applications, studying reinforcement details and ARE code–based practices. They gained exposure to structural drawings, site execution, and discussions on design and safety considerations. Overall, the visit enhanced technical skills and understanding of project management and professional responsibilities in civil engineering.

4. An industrial visit was organized for M. Tech Construction Engineering and Management students to the C2 Mumbai High-Speed Rail (Bullet Train) Project. The visit, scheduled on 16 October 2025, involved 38 students and 3 faculty members. It provided valuable exposure to large-scale infrastructure planning, construction practices, and project management aspects of high-speed rail systems.

Department of Petroleum

The viist to Baker Hughes,Powai, Mumbai by Dr Shailendra Naik,Prof Samarth Patwardhan,Dr Ajendra Singh,Dr Soubir Das Dr Kumar Abhijeet Raj and students of department of petroleum engineering

Department of Polytechnic

Industry visits provide students with valuable hands-on experience and insights into real-world applications of their academic knowledge. These visits allow students to observe cutting-edge technologies, interact with industry experts, and understand the practical challenges and solutions in various sectors

Around 200 students of SYCSE and SYAIDS visited PHN Technology, Viman Nagar, Pune on 22/08/2025.

Students of TYME-RA visited Horizon Packtech, Aarvi on 25/08/2025.

Students of TYME-RA visited Nextech Automation Solution, Aarvi on 25/08/2025.

Students of TYCE and faculty members visited Sewage Treatment Plant ,Sihgad Road on 09/09/2025

Students of SYCE and faculty members visited Railway Station ,Pune on 10/09/2025

Students of SYCE and faculty members visited Building Construction Super Structure,Dahanukar Colony,Pune on 19/09/2025

Students of SYCE and faculty members visited the Building Construction Sub. Structure ,Bhelke Nagar on 19/09/2025

SY and TY students of ECE-AIML visited PHN Technology,Viman Nagar,Pune on 19/09/2025

A visit to Koyna Dam was organized for 21 TYCE students along with 2 faculty members on 23/09/2025

A visit to the Funicular Ropeway, Kalyan was organized for 28 TYCE students along with 2 faculty members on 25/09/2025.

Alumni Engagement

Department of Civil

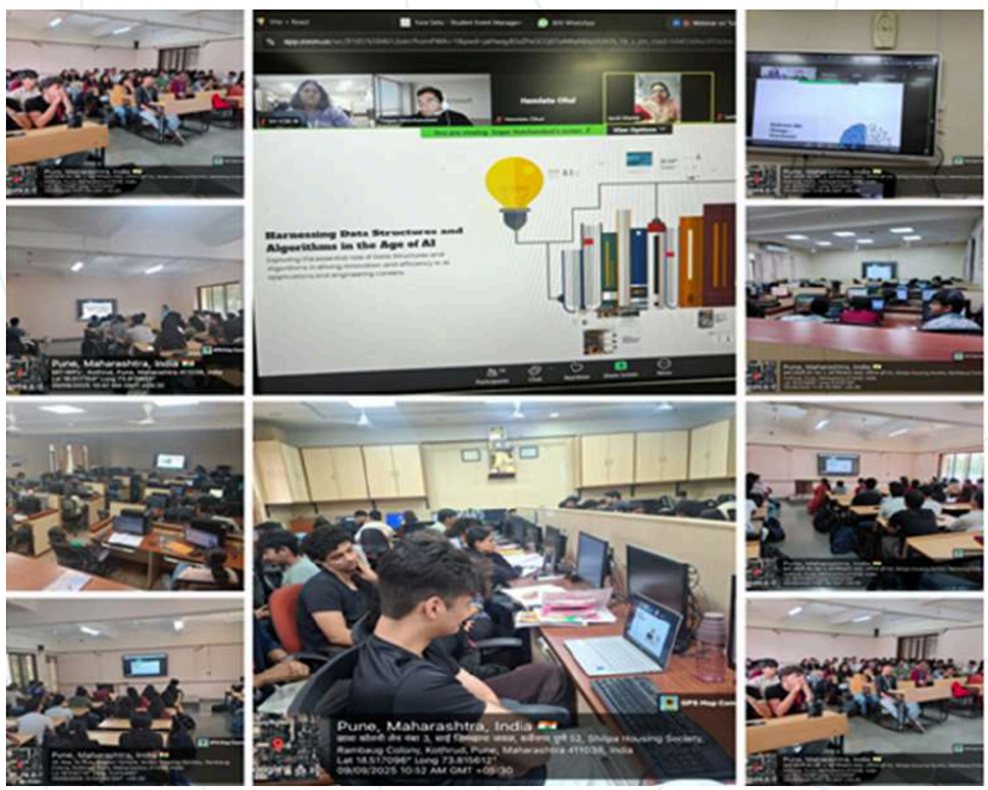
A guest lecture was conducted by the Department of Civil Engineering on 09 April 2025 for First Year B.Tech and M.Tech students. The session was delivered by Mr. Vedang Patil, Founder of Innovation Hub & Ionosphere and an alumnus of MIT-WPU. The lecture focused on preparation for foreign studies and current industry demands in civil engineering. The speaker shared insights on higher education pathways, global opportunities, and evolving industry trends. The interactive session was highly informative and beneficial for students' academic and career planning.

A GATE Exam Preparation Session was organized on 22nd September 2025 for final-year B. Tech students to guide them in systematic exam preparation. The session was conducted by Mr. Hajarathvali Shaik, an alumnus of MIT-WPU and experienced GATE mentor, who explained the exam structure, syllabus, and marking scheme in detail. Effective preparation strategies, time management techniques, and the importance of conceptual clarity were emphasized. Students were guided on the use of reference books, online resources, and mock tests. The interactive session boosted students' confidence and motivation towards achieving success in the GATE examination.

Facilitation of the expert by the faculty member Dr. Ganesh Ingle

Department of Polytechnic

The webinar on "Software Engineering in an AI World" was organized on 9th September 2025 at 10:30 AM IST for SY-CSE, SY-AIDS, TY-CSE and TY-AIDS students by the Computer Engineering Department. The session was delivered by our esteemed alumnus, Mr. Sagar Hotchandani, Principal Software Engineering Manager at Microsoft Corporation, Greater Seattle. The session was highly insightful, and the attendees found it to be extremely valuable.



Achievement & Milestones

Department of Petroleum

The Final Year (Petroleum Engineering) Moharjyoti Chakraborty has been awarded 3rd place at SPE 2025 South Asia Virtual Student Paper Contest award organized by Society of Petroleum Engineers.

Department of Polytechnic

Students from the Polytechnic department achieved remarkable milestones across various competitions

Amey Ajgar from TYMRA to get the first position in this MaTPO competition. He won GD Idol competition amongst almost all engineering students from across Maharashtra

Maha Association of Training & Placement Officers (MaTPO)

Congratulations to

Winners of "MaTPO Group Discussion (GD) Idol - 2025"

MaTPO has conducted an online Group Discussion (GD) contest from 13-08-2-25 to 26-08-2025 for First Year to Final Year Maharashtra students of BE / B.TECH / ME / M.TECH / DIPLOMA / BCA / MCA / BCS / MCS / BBA / MBA / MMS / B.Sc / M.Sc / D.Pharm / B.Pharm / M.Pharm / BA / MA / B.Com / M.Com / Other Courses in 5 Rounds.

Following are the WINNERS of MaTPO Group Discussion (GD) Idol - 2025 :

Rank	Student Name	College	Course	Branch	Batch	Prize
1	Amey Ajgar	MIT World Peace University, Pune	Diploma	Mechanical	2026	Mobile
2	Hrishikesh Shahane	D.Y. Patil College of Engineering & Technology, Kolhapur	B.Tech	CS (AI-ML)	2026	Tab
3	Payal Chavan	MIT ADT University, Pune	B.Tech	CSE	2026	FitBit
4	Vaidehi Deshmukh	D.Y. Patil College of Engineering, Akurdi, Pune	B.Tech	AI&DS	2026	Backpack
4	Shreeya Holikatti	N.K. Orchid College of Engineering & Technology, Solapur	B.Tech.	CSE	2026	Backpack

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MaTPO Council : https://tinyurl.com/MaTPOCouncil

Amey Ajgar from TYMRA to got the Best paper presentation award in conference HSWTech 25 organized by Department of Electrical and Electronics Engineering,MITWPU,Pune.

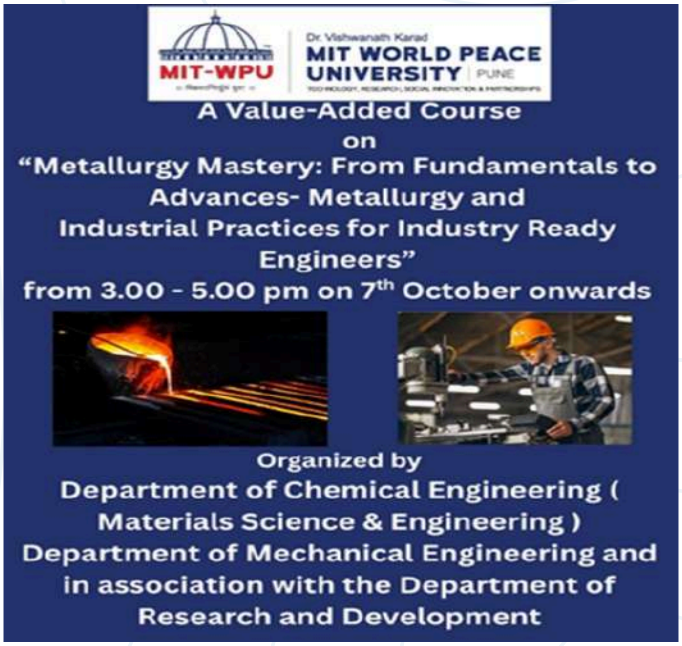
IIIT Bhubaneswar has organized a Hackathon for Maharashtra teams, the 1st Round will be conducted in hybrid mode on 27th September 2025, co-hosted by the SIT ACM Student Chapter. The event is primarily organized by IIIT Bhubaneswar. From this Maharashtra round, two winning teams will be selected to participate in the Final Round at IIIT Bhubaneswar from 7th to 9th November 2025.Our team of students from Polytechnic, Team CodeForge consisting of members Kaustubh Pandey, Vivaan Mathur, Shrinath Nawale and Siddharth Mane from SYCSE and TYCSE-AIDS secured first place out of 95 teams all over Maharashtra in the Rewind and Recode Hackathon hosted by IIIT Bhubaneswar and Symbiosis Institute of Technology, Pune. The event was held in Symbiosis University and our team was selected to go to the national round to be held in IIIT Bhubaneshwar.Mane from SYCSE and TYCSE-AIDS secured first place out of 95teams all over Maharashtra in the Rewind and Recode Hackathon hosted by IIIT Bhubaneswar and Symbiosis Institute of Technology, Pune. The event was held in Symbiosis University and our team was selected to go to the national round to be held in IIIT Bhubaneshwar.

Department of Chemical Engineering

Contribution to University Activities

- Dr. Anand D. Kulkarni served as a core organizing committee member for the World Technology Summit 2025 (WTS 2025) held on 5-6 Nov 2025.
- Dr Dinesh Bhutada represented MIT-WPU in ICSK-IIK Higher Education Fair 2025 in Kuwait
- Dr Dinesh Bhutada represented Division 1 in Career Guidance Faire in Jalgaon on 30th Nov 2025

Value added Course on 'Metallurgy Mastery: From Fundamentals to Advances Metallurgy and Industrail Practices for Industry Ready Engineers' from 07-10-2025 for SY and TY MSE students.



Department of Bio Engineering

Team Udajana (Anshul Laxane, Ayush Yadav, Parth Patil, Ārya Yādav, Sakshi Desai, and Aditya Patel) has successfully won the internal rounds of Smart India Hackathon (SIH)and qualified for the SIH 2025 Main Round! Out of 500+ participating teams, only 90 were selected — and Team Udajana are one of them. They were competing under the sector of Clean & Green Technology, focused on the topic: Biohydrogen Production Using a Biological Pathway, registered under the problem statement SIH25111. This journey wouldn't have been possible without the invaluable guidance of our mentors: Dr Sagar Kanekar Sir, Dr. Anand D Kulkarni Sir, Dr. Bharat Kale Sir. A special thanks to Vishal Pillai Sir for mentoring us throughout and helping refine our presentations with professional excellence. We are also grateful for the constant support and encouragement from Dr. Sharbani Kaushik, Dr. Vikrant Gaikwad Sir (Dean Academics), Dr Sidharth Chakrabarti Sir (Dean SOET).

Research output workshop was organized by Research and Development group and faculties presented their research work in front of foreign research experts and suggested inputs on the future work. Bioengineering faculty Dr. Sagar Kanekar presented research work from Bioengineering department.

Placements & Career Progression

Campus Placement Highlights:
Department of EEE-

Sr. No.	Name of the placed student	Designation	Name of the Company	CTC	Photograph of student	Company logo
1	Rudra Pratap Singh	ZS associates	Business Technology Solutions Associate	14.5		
2	Devyani Sisodia	ZS associates	Business Technology Solutions Associate	14.5		
3	Aishwarya Godse	ZS associates	Business Technology Solutions Associate	14.5		
4	Clera Jack Tribhuvan	Varroc Engineering Limited	Trainee	12.5		
5	Kushagr Bansal	ProcDNA Analytics Private Limited	Technology Analyst	11.74		
6	Krish Sachin Menon	Semtech	Software Developer Intern /QA Engineer Intern /Hardware Engineer Intern	11		
7	Maitreyee Choudhary	Semtech	Software Developer Intern /QA Engineer Intern /Hardware Engineer Intern	11		

8	Isha Gour	Philips	R&D H/W Electrical Engineer	11		
9	Jha Rohan	ZS Associates	Business Technology Solutions Associate	13.68		
10	Isha Gour	ZS Associates	Business Technology Solutions Associate	13.68		

Department of Petroleum

Sr. No.	Name of the placed student	Designation	Name of the Company	Company logo
1	ATHARV JADHAV	Petroleum engineer	CHEVRON	
2	Akif AHMED	Facility engineer	CHEVRON	

Department of Chemical Engineering

Sr No.	Faculty Name	Qualification	Areas of interest	Date of Joining
1	Dr Kiran P. Shejale	BTech (Shivaji University), MTech (IIT Roorkee), PhD (IIT Jodhpur)	Nanomaterials & Functional Coatings, Energy Storage & Conversion, Catalysis, Electrochemical Systems & Smart Interfaces and Water Treatment & Environmental Sustainability	03rd Feb 2025
2	Dr Anjali Meshram	BTech (LIT Nagpur) MTech (VNIT Nagpur) PhD (BITS Pillani K K Birla Goa Campus)	Water remediation, membrane catalysis, nanomaterials, biomass valorization	13th Feb 2025
3	Prof Anant Srivastava	B.tech(NIT Durgapur), PhD thesis Submitted (IIT Kanpur)	Nanomaterials, Raman Spectroscopy, Carbon Materials,Enviornment, and Charge Transfer	11th August 2025

Department of Chemical Engineering

SR. No.	Name of the placed student	Designation	Name of Company	CTC	Photograph of student	Company logo
1	Sakshi Sudhir Kinage	Associate Engineer Intern	Johnson and Johnson, Iziel Healthcare, Jyesta entity	6.5 LPA		
2	Riya Vipin Tyagi	Associate consultant Level-1	GRG Healthcare and FMI Consultants	4.61 LPA		
3	Swarali Vijay Nikam	Associate consultant Level-1	Izieil Helathcare and FMI Consultants	4.61 LPA		
4	Arpita Deep	Associate consultant Level-1	FMI Consultants	4.61 LPA		
5	Shriya Prakash Pote	Summer Internship	Jonson & Jonson	4.44 LPA		
6	Iravati Dhembare	Customer Service Representative.	Medi Transcare Pvt. Ltd.,	4 LPA		

Workshops, Expert Lectures/Webinars & Conferences:

Workshop

1.A hands-on training program on Drone Technology was conducted by Mr. Krish Mehta from the Drone Club, MITWPU, from 21/08/2025 to 25/08/2025. The program provided participants with practical exposure to drone design, operation, and real-world applications.

2. A CSSC, New York (USA)–accredited Lean Six Sigma Yellow Belt program in Process Optimization with Design of Experiments was conducted by Mr. Santosh Awasarkar, Senior Consulting Partner, TNS, Singapore, from 28/08/2025 to 29/08/2025. The program focused on improving process efficiency through data-driven problem-solving techniques.

3.A practical session on Two-Wheeler Repair and Maintenance was conducted at Yash Service Center on 09/09/2025. The session provided hands-on exposure to routine servicing, troubleshooting, and maintenance practices.

4.A workshop on Arduino Uno with Project Development was conducted from 26/09/2025 to 27/09/2025. The program focused on hands-on learning, enabling participants to design, program, and develop basic Arduino-based projects.

- MNRE COE for Hydrogen Production and Storage was selected for defense presentation.
- Information regarding inclusion of NPTEL courses in upcoming curriculum according to NEP was given by faculty Dr. Sagar Kanekar to S Y B Tech Bioengineering students in online mode.

- Overview of research work by students was given by third year bioengineering students to upcoming first year students taking admission for 2025-29 batch.

Expert Lectures

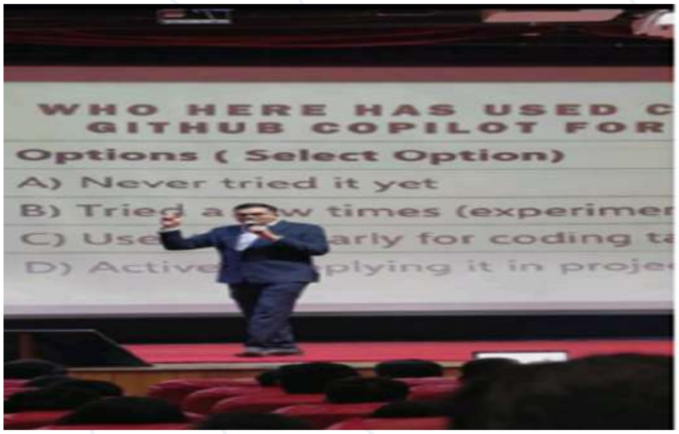
A guest lecture on Drone Technology was delivered by Dr. Ajit Kumar on 03/09/2025. The session provided insights into drone fundamentals, applications, and emerging trends in the field.

A technical session on PLC Automation and Control Systems was conducted on 17/09/2025 by Mr. Nilesh Wagh from Electrosoft Systems. The session focused on the fundamentals, applications, and industrial relevance of PLC-based automation.

A technical session on Advances in Welding was conducted on 15/09/2025 by Shri Chandrashekhar Vaidya. The session highlighted recent developments, modern techniques, and industrial applications in welding technology.

A guest lecture on Blockchain Technology was delivered by Dr. Ram Kumar Solanki on 26/09/2025. The session provided an overview of blockchain fundamentals, applications, and its impact on modern industries.

A session on Prompt Engineering was conducted on 27/09/2025 by Mr. Surendra Panpaliya. The talk focused on effective prompt design techniques and their applications in AI-driven systems.



A guest lecture titled “Innovate, Adapt and Succeed: Preparing Engineers for the Future” was delivered by Mr. Anant Joshi, Business and Technology Consultant, on 30/09/2025. The session emphasized future-ready skills, innovation, and adaptability in the evolving engineering landscape.

Campus Events,Activities & Clubs

Department of EEE

a) The Semiconductor Conclave 2025 was organized by the Department of Electrical and Electronics Engineering, School of Engineering and Technologyon 29th July 2025 at MIT-WPU, Pune that focused on India's strategic rise in the semiconductor and electronics ecosystem. Key experts Dr. Satya Gupta, President VLSI Society of India, Dr. Pannerselvam Madangopal, CEO, Meity Startup Hub, Govt. of India and Mr. H. S. Jatana noted semiconductor industry leader highlighted India's opportunities in fabless design, workforce alignment, and gradual development of indigenous fabrication

b) The IEEE International Conference on, "Sustainable Technologies for Humanity and Smart World (HSWTech-2025) has concluded successfully at MIT World Peace University. The conference is organized by the Department of Electrical and Electronics Engineering, School of Engineering and Technology in association with IEEE Pune Section.
Under HSWTech-2025 the pre-conference workshops were held at MIT-WPU, Pune on topics of VLSI & FPGA Tool Flow Unlocked: Insights, Wearable Microstrip Antenna and Tiny ML to Cloud.
The inaugural session of HSWTech-2025 was graced by the presence of Mr. Abhijit Atale (IBM) Dr. Subhra Kanti Das (Thales Research & Technology), Mr. Sunil Kharate (Broadcom), Mr. Nikhil Datar (LTIMindtree). The keynote speakers Ms. Sai Jadhav, Senior Engineer at Qualcomm delivered insights on cutting-edge autonomous driving technologies along with Mr. Shripad Kale, Director R&D, Schneider Electric shared expertise on innovation and sustainability in industrial automation.
Dr. Sunil Gorantiwar, Head of Agricultural Engineering, Mahatma Phule Krishi Vidyapeeth, Rahuri delivered an insightful session on sustainable agricultural technologies, emphasizing innovations that support ecological balance and enhance productivity.

A pre-conference workshop under the international Conference, HSW Tech 2025 on the topic TinyML to Cloud was conducted on 22nd August 2025 by Mr. Arunkumar Nair, an expert in embedded AI systems from Canspirit AI for the students and staff of DoEEE. He highlighted the rapid growth of TinyML and its applications in areas such as IoT, healthcare, robotics, and smart devices. The importance of deploying AI efficiently on constrained devices was emphasized, along with the role of cloud platforms like Edge Impulse in simplifying the development process. Dr. Manisha Kowdiki coordinated the workshop.configured their environments for TinyML development, gained familiarity with Edge Impulse tools, built and trained a functional image classifier,experienced AI-assisted labelling techniques,deployed and tested their models on actual hardware.

RIDE -2025

On 14 August 2025, SkinTwin app that detects facial skin problems and recommends products,developed byNikita Kulkarni, Saee Sukale, Rakshita Bajaj of second year ECE AIML ,DoEEE students received ₹1,00,000 funding for prototype development from investor Suketu Sanghvi under a flagship entrepreneurship initiative of RIDE 2025, organised in collaboration with the Wadhwani Foundation.

Department of Civil

A guest lecture was organized for B. Tech Civil Engineering students in offline mode at VK204. The session was delivered by Mr. Devesh Jain from Vincon Valuations on 30 October 2025. He discussed approaches in valuation and career opportunities, offering valuable industry insights and practical guidance to students.

MIT-WPU, in collaboration with AUM–LAI Chapter Pune, successfully hosted a One-Day International Conclave on Urban Planning and Land-Use Science. The event featured leading national and international experts who shared insights on land-use policy, climate-resilient planning, and sustainable urban development. With participation from senior administrators, policymakers, and global scholars, the conclave emphasized integrated approaches to land economics, governance, and social equity. The discussions reaffirmed MIT-WPU's role as a hub for innovation, research, and global partnerships in civil engineering and urban management.

The guest lecture on "Role of Intelligence Quotient in Construction Management" provided a clear and engaging overview of how IQ contributes to effective project management in the construction industry. The speaker highlighted how high IQ enhances problem-solving, decision-making under pressure, planning, and conflict resolution—skills that are essential in complex construction projects. Realistic project scenarios helped connect theory with practice, making the session informative and relatable. Overall, the lecture effectively emphasized the importance of cognitive abilities alongside technical knowledge in shaping successful construction managers.

Department of Petroleum

The field visit to Gargoti Museum organized by SPG,MIT WPU chapter accompanied by Dr Rahul Joshi,Prof Rajib Kumar Sinharay,Dr Kunal Modak,Dr Ajendra Singh

The clean drive organized by SPE,MIT WPU Chapter for UG&PG Petroleum Engineering students accompanied by Prof Samarth Patwardhan, Dr Minal Deshmukh, Dr Ajay Sahu and Dr Kumar Abhijeet Raj

The inauguration of Research scholar lab(BJ 308A) at Department Petroleum Engineering by Dean,SOET accompanied by faculties of department.

Department of Chemical engineering

Metallurgy Day Celebration Program

Metallurgy Day was celebrated for the students of BTech Materials Science & Engg on 14th November 2025 .

Department of Bio-Engineering

- **The Third-Year Bioengineering students of MIT-WPU had an absolutely thrilling and eye-opening experience at the Global ED Expo 2025, organised by TC Global at the Sheraton Grand, Pune, on 13th September 2025.**
- **The event buzzed with energy as top universities from the UK, USA, and Ireland came together under one roof, eager to connect with students and share all that makes their institutions stand out. From cutting-edge research opportunities and diverse course offerings to insights into campus life and global scholarships, every stall had something new and exciting to discover!**
- **The representatives were incredibly enthusiastic and approachable, patiently guiding us through everything from choosing the right programs to understanding the visa application and admission process. Their step-by-step explanations made the idea of studying abroad feel less daunting and more like an achievable dream. It was amazing to see how education can truly open doors to the world!**

- The entire experience was a perfect mix of fun, inspiration, and learning. Walking through the expo, chatting with representatives, and collecting brochures felt like taking the first steps toward our global journeys. The event left us all buzzing with motivation and a clearer sense of direction for our future studies abroad.

- Sincere gratitude to Dr.Shraddha Kulkarni Ma'am, Co-ordinator of this networking event, for providing this opportunity for the students.

- Heartfelt appreciation to Dr. Sharbani Kaushik, PhD Ma'am, Program Director Department of Bioengineering, Dr. Sagar Kanekar Sir, faculty coordinator of SBE, Dr Sidharth Chakrabarti Sir, Dean of SOET, Dr. Vikrant Gaikwad Sir, Dean of Academics; for ensuring the event's success through their support.

- SBE organized 'Team Meet Up' with 'Partex AI' team and focusing on cutting edge AI applications with regulatory and research insights with industry leaders.

AIChE SBE organized Gurupournima program and all the teaching, non-teaching staff were felicitated. A cultural program was performed by the students.

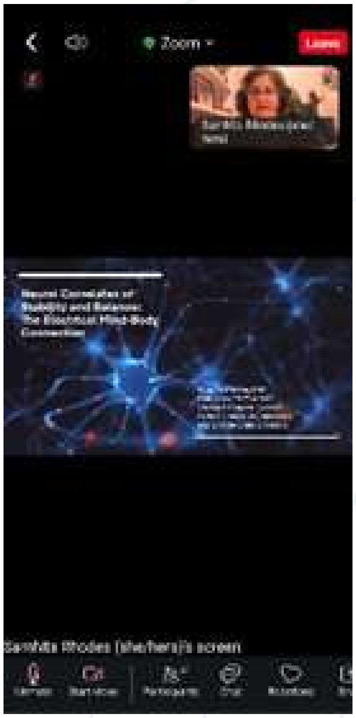
Launching of The Neuro-nexus Series, an initiative by Society for Biological Engineers (SBE), MIT-WPU, designed to bring our students closer to industry leaders, researchers, and global professionals in the field of Biomedical Engineering was done on 23rd August. An exclusive online interaction with Dr. Samhita Rhodes (Professor of Biomedical Engineering, Grand Valley State University (GVSU), Michigan, USA) was organized online.

Session Highlights:

- Insights into Biomedical Signal Processing Research
- Broader perspectives on the Role of Bioengineers
- Opportunities for Higher Studies at GVSU
- Summer Experience Program at GVSU (2-week project)
- Joint Projects & Thesis Co-supervision with GVSU faculty

- This gave an invaluable opportunity for students to learn, network, and explore global avenues in Biomedical Engineering.

- We thank our Programming Director Dr. Sharbani Kaushik, PhD Ma'am and Faculty advisor of SBE Dr Sagar Kanekar Sir, for giving us this wonderful oppurtunity to connect with our guest!



Ninox organized cleanliness drive and tree plantation drive with the club members in and around Pune. Students from S Y and T Y Bioengineering actively participated in the drive to help the sustainability goals set by MIT WPU university.

MIT WPU SBE students chapter presented the club activities at Shubharambh 2025 on 15th July 2025

Student & Faculty Corner

Department of EEE

a) Shreya Aigalikar, student of Department of Electrical and Electronics Engineering, MITWPU won the first prize for the project titled "Raag Vision - Decoding Hindusthani Classical Music with Machine Learning" under poster competition at National Scientist Round Table Conference.

b)Team "Eco-Campus" mentored by Dr Anuradha C. Phadke, secured the 1st Prize at the Youth for Earth Conclave, organized by the Mobius Foundation and Climate Change Reality Project, India -South Asia at the "India Habitat Centre, New Delhi on 17th and 18th September 2025.

The Second year B. Tech ECE, DoEEE MITWPU student Akshita Saxena has made the university proud by securing 3rd Prize at the state-level elocution competition – Voice of Devendra 2025. She was felicitated by Hon. Ravindra Chavan, Cabinet Minister.

The Second year B. Tech ECE, DoEEE MITWPU student Jaskiran Kaur Dhami won three gold medals in 12th West Zone Shooting Championship in Shotgun events at M.P. Shooting Academy, Bhopal Madhya Pradesh organised from 18th to 21st September, 2025.

Department of Civil Engineering

The Faculty Corner highlights distinguished achievements of faculty members who completed industry internships from May to June 2025, strengthening academia–industry collaboration. It showcases faculty who engaged with leading construction and infrastructure organizations, gaining practical exposure to emerging technologies and management practices. The section also features a Research Internship by Dr. Meera (Civil Engineering), recognizing her contribution to advancing knowledge in her specialization. Through these internships, faculty enhanced curriculum relevance, initiated collaborative projects, and enriched student learning with real-world insights.

Overall, the Faculty Corner celebrates continuous professional growth, innovation, and impactful engagement with industry and research communities.

Department of Polytechnic

Sohan Rane
SY CSE(AI-DS),Department of Polytechnic

AI and Data Science are engineering a paradigm shift in the financial ecosystem, fusing computational intelligence with economic logic. Machine learning architectures like decision trees, gradient boosting models, and deep neural networks process terabytes of transactional data to identify anomalies and optimize operations. Natural language processing enables automated report generation and customer interaction, while reinforcement learning dynamically fine tunes investment and credit strategies based on evolving market signals.

In modern banking infrastructure, data pipelines integrate structured and unstructured data through ETL frameworks and distributed storage systems like Hadoop and Spark, ensuring scalability and fault tolerance. Predictive analytics models then evaluate customer creditworthiness using multidimensional features beyond traditional credit scores, such as behavioral metrics and real time financial flows. This drastically reduces default probabilities and operational risk.

In the loans domain, explainable AI frameworks ensure transparency by revealing how each feature contributes to credit decisions, satisfying both ethical and regulatory standards. Smart contract systems powered by blockchain further secure loan agreements with cryptographic integrity and automated enforcement.

Ultimately, AI and Data Science transform finance from a rule based structure into an adaptive, learning system. They empower banks to make precision driven decisions, optimize liquidity, and deliver hyper personalized services, setting the foundation for an autonomous, data centric financial future.

Department of Bio Engineering

Team ’BioRhodeon’ working on Biosensitized Solar Cell presented their work in Smart India Hackathon and cleared third round and are in waitlisted round of final round.


MIT WORLD PEACE UNIVERSITY

ले चल

MIT-WPU

SMART INDIA HACKATHON 2025

SIH

(University Internal)

Certificate Of Participation

THIS CERTIFICATE IS PROUDLY PRESENTED TO

Asya Shingade

for taking part in MIT-WPU SMART INDIA HACKATHON [SIH] University Internal 2025, at Dr. Vishwanath Karad MIT World Peace University, Pune

Dr. Prasad Khandekar, Chief Academic Officer, MIT-WPU

Prof. Dr. Milind Pande, Convener SIH, Pro-Vice Chancellor, MIT-WPU

Prof. Dr. R. M. Chitnis, Vice Chancellor, MIT-WPU

Shri. Rahul V. Karad, Executive President, MIT-WPU

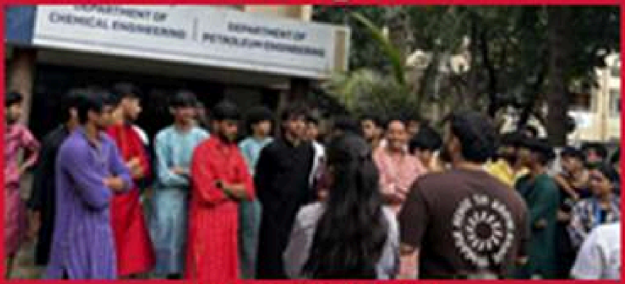
AVINYA

Department of Petroleum Engineering

18th Aug 2025

Orientation of IADC MIT SC

25th Aug 2025



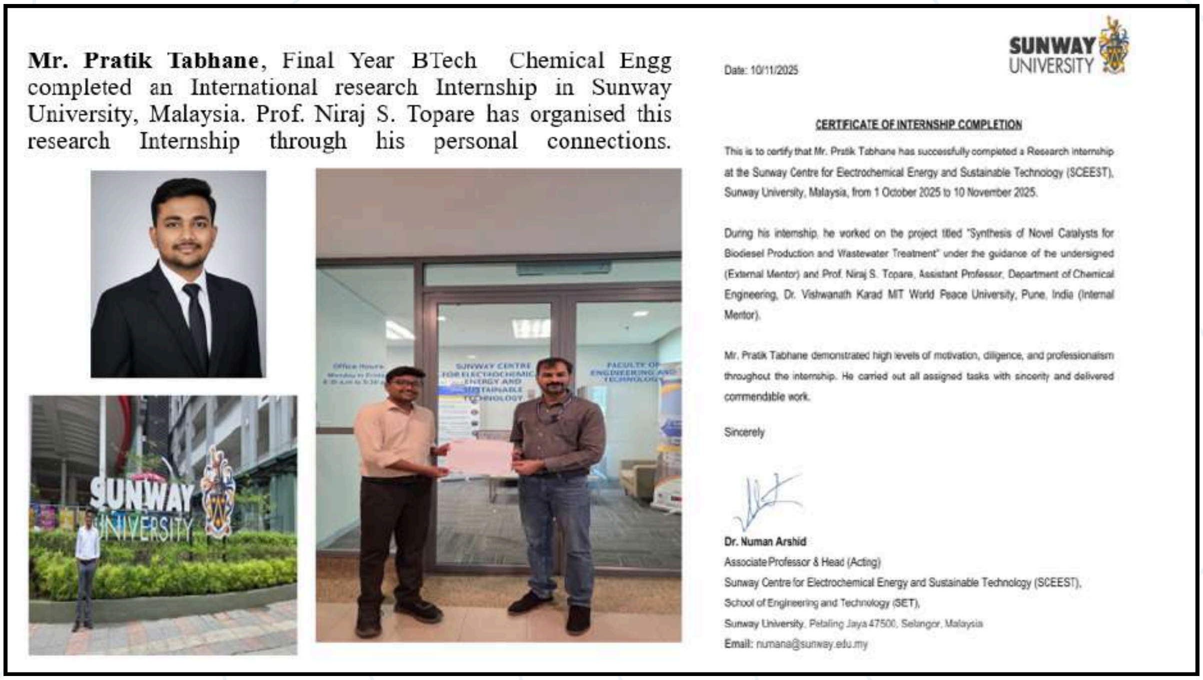
Treasure Hunt

03rd Sept 2025

Rig Cube 2.0

Department of Chemical Engineering

Mr. Pratik Tabhane, Final Year BTech Chemical Engg completed an International research Internship in Sunway University, Malaysia. Prof. Niraj S. Topare has organised this research Internship through his personal connections.


Mr. Pratik Tabhane, Final Year BTech Chemical Engg completed an International research Internship in Sunway University, Malaysia. Prof. Niraj S. Topare has organised this research Internship through his personal connections.

SUNWAY UNIVERSITY

Date: 10/11/2025

CERTIFICATE OF INTERNSHIP COMPLETION

This is to certify that Mr. Pratik Tabhane has successfully completed a Research internship at the Sunway Centre for Electrochemical Energy and Sustainable Technology (SCEEST), Sunway University, Malaysia, from 1 October 2025 to 10 November 2025.

During his internship, he worked on the project titled "Synthesis of Novel Catalysts for Biodiesel Production and Wastewater Treatment" under the guidance of the undersigned (External Mentor) and Prof. Niraj S. Topare, Assistant Professor, Department of Chemical Engineering, Dr. Vishwanath Karad MIT World Peace University, Pune, India (Internal Mentor).

Mr. Pratik Tabhane demonstrated high levels of motivation, diligence, and professionalism throughout the internship. He carried out all assigned tasks with sincerity and delivered commendable work.

Sincerely

Dr. Numan Arshid
Associate Professor & Head (Acting)
Sunway Centre for Electrochemical Energy and Sustainable Technology (SCEEST),
School of Engineering and Technology (SET),
Sunway University, Petaling Jaya 47500, Selangor, Malaysia
Email: numana@sunway.edu.my

Media Recognition

Department of Civil Engineering

hindustantimes

Gadkari opens India's first tunnelling centre at MIT-WPU

PUNE: Union minister for road transport and highways, Nitin Gadkari, inaugurated the International Workshop on 'Sustainable Tunnelling for Better Life' at MIT World Peace University (MIT-WPU). Organised in collaboration with ITA-CET, the two-day event attracted experts from India, Europe, United Kingdom and United States of America.

A key highlight was the launch of the Centre of Excellence for Tunnelling and Underground Construction, India's first such facility, developed in partnership with Tata Projects Ltd. and Sandvik.

The centre features advanced tunnel monitoring and drilling and blasting laboratories to promote research and training. HTC

The Indian EXPRESS
JOURNALISM OF COURAGE

Tunnel projects worth Rs 3 lakh cr in next 10 years in India: Gadkari

Padma Vibhushan Dr Vijay Kelkar receives the Punyabhushan Puraskar from Union Minister Nitin Gadkari at Balgandharva Rangamandir in Pune on Monday.

THE TIMES OF INDIA

Govt set to spend ₹3L crore on tunnels in next 10 yrs: Gadkari

Pune Likely To Receive Infra Projects Worth ₹50,000 Crore

Scope for industry to refurbish tunnelling machines

Tunnels to be made in Build Operate Transfer model

Relaxing eligibility requirements impacted quality

Dharmadhikari receives lifetime achievement award

Centre of excellence for road construction

पुण्य नगरी

पुढील १० वर्षात २.५ ते ३ लाख कोटींच्या भुयारी मार्गांची उभारणी : गडकरी

Pune Edition
Jun 24, 2025 Page No. 11
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महाराष्ट्र टाइम्स

'बोगद्यांसाठी तीन लाख कोटी'

नितीन गडकरी यांची माहिती

'५० हजार कोटींची पुलाची कामे सुरू'

'पुणे-सोलापूर रस्ता, चाकण-शिक्रापूर रस्ता, पुणे-नाशिक रोड अशा पुण्यासाठी ५० हजार कोटी रुपयांचे पूल बांधण्याचे काम सुरू आहे. पुणे-छत्रपती संभाजीनगर दरम्यानचा प्रवास दोन तासांपर्यंत कमी होईल. त्यासाठी बीओटी मॉडेलमध्ये रस्त्याची निर्मिती करण्याचा विचार करीत आहोत,' असे नितीन गडकरी यांनी सांगितले.

म. टा. प्रतिनिधी, पुणे

'पुढील १० वर्षांत भारतात बोगदे तयार करण्यासाठी तीन लाख कोटी रुपये खर्च केले जातील,' अशी माहिती केंद्रीय रस्ते वाहतूक व महामार्गमंत्री नितीन गडकरी यांनी सोमवारी दिली. बाजारभावापेक्षा ४० टक्के कमी दराने कंत्राटदार काम करीत असल्याने, अनेक बड्या कंपन्यांनी आमच्यासाठी काम करणे बंद केले आणि वाईट कंत्राटदारांना काम मिळाले. या कंत्राटदारांना तुरुंगात टाकायचे आहे, असेही गडकरी यांनी स्पष्ट केले.

एमआयटी वर्ल्ड पीस युनिव्हर्सिटीतर्फे (एमआयटी डब्ल्यूपीयू) 'सस्टेनेबल टनेलिंग फॉर बेटर लाइफ' या विषयावरील कार्यशाळेच्या उद्घाटनप्रसंगी गडकरी बोलत होते. या वेळी 'आंतरराष्ट्रीय टनेलिंग असोसिएशन'चे माजी अध्यक्ष अरनॉल्ड डिक्स, 'एमआयटी डब्ल्यूपीयू'चे संस्थापक डॉ. विश्वनाथ कराड, कार्याध्यक्ष राहुल कराड आदी उपस्थित होते.

या कार्यशाळेत भारत, युरोप, ब्रिटन आणि अमेरिका येथील तज्ज्ञ सहभागी झाले असून, 'इंटरनॅशनल टनेलिंग असोसिएशन'च्या सहकार्याने हे आयोजन केले आहे. कार्यक्रमात सेंटर ऑफ एक्सलन्स फॉर टनेलिंग अँड अंडरग्राउंड कन्स्ट्रक्शनचे उद्घाटन करण्यात आले. हे केंद्र सँडविक आणि टाटा प्रोजेक्ट्स लि. यांच्या सहकार्याने उभारले असून, यात टनेल मॉनिटरिंग लॅबरिटरी आणि ड्रिलिंग अँड ब्लास्टिंग यांचा समावेश आहे.

गडकरी म्हणाले, 'बोगदे केवळ प्रवासासाठी नाहीत, तर सुरक्षा, पर्यावरणीय संवर्धन आणि टिकाऊ विकासासाठीही अत्यंत गरजेचे आहेत. आपण या क्षेत्रात स्वयंपूर्ण होण्याचा निर्धार केला पाहिजे. टनेल बांधणीचा खर्च कमी करताना गुणवत्ता कायम ठेवावी लागेल. यासाठी सीएनजी, इथेनॉल, हायड्रोजन आणि इलेक्ट्रिक इंधनांचा वापर वाढवणे गरजेचे आहे. जुन्या टनेलिंग मशिनचे पुनर्निर्माण करणे, युरोपातील ऑस्ट्रिया, नॉर्वे आणि स्पेनमधून वापरलेल्या यंत्रणा आयात करणे आणि पुढे जाऊन भारतातच त्या उत्पादनात आणणे, ही दिशा असली पाहिजे.'

पुढारी

शहरातील वाहतूक करणार भूमिगत मार्गाने

देशभरात ३ लाख कोटी रुपयांचे बोगदे उभारणार : नितीन गडकरी यांची माहिती

TheHitavada

Nagpur City Line | 2025-06-24 | Page- 6
ehitavada.com

Plan to build tunnel projects worth Rs 2.5 to Rs 3 lakh crore: Nitin Gadkari

■ Business Reporter

UNION Minister for Road Transport and Highways Nitin Gadkari, on Monday inaugurated the International Workshop on 'Sustainable Tunnelling for Better Life' at MIT World Peace University (MIT-WPU). The two-day event organised in collaboration with the International Tunnelling and Underground Space Association's Committee on Education and Training (ITA-CET) has brought together global experts from India, Europe, the UK, and the US.

During his address, Gadkari said: "India is entering a golden era of infrastructure development, with tunnels playing a crucial role in connectivity, safety, and sustainability. Over the next 10 years, we plan to build tunnel projects worth Rs 2.5 to Rs 3 lakh crore. To make this a reality, we must reduce construction costs without compromising quality. That means using new technologies and sustainable fuels like CNG, ethanol, hydrogen, and electric alternatives to diesel." Arnold Dix mentioned, "This Centre of Excellence is of global importance, as it addresses the disconnect between engineering expertise and the practical skills."

Dr Vishwanath Karad, founder MIT-WPU said: "Despite advancements in education, science, and technology—from Facebook to AI—we still lack tools that truly foster peace."

Speaking at the event, Dr Rahul Karad, Executive President of MIT-WPU said: "The inauguration of the two laboratories marks a significant step forward in fostering academia-industry collaboration. With the generous support of Tata and Sandvik, contributing equipment valued at nearly Rs 2 crore, we are strengthening an ecosystem where research, innovation, and national development align."

लोकसत्ता
लोकमान्य लोकशक्ती

पुढील दशकभरात अडीच ते तीन लाख कोटींचे बोगदे प्रकल्प

केंद्रीय रस्ते वाहतूक आणि महामार्ग मंत्री नितीन गडकरी यांची माहिती

लोकसत्ता प्रतिनिधी

उत्कृष्टता केंद्राची स्थापना

'काही कंत्राटदार, टोलवाल्यांना तुरुंगात टाकणार'

BW BUSINESSWORLD
Tuesday | Jun 24 2025 | 03:05:07 PM

Home / Economy / India Eyes Rs 3 Lakh Cr Tunnelling Push: Nitin Gadkari

India Eyes Rs 3 Lakh Cr Tunnelling Push: Nitin Gadkari

BW Online Bureau | Jun 24, 2025

India is gearing up for a massive expansion in tunnel infrastructure, with projects worth Rs 2.5 to Rs 3 lakh crore planned over the next decade, Union Minister for Road Transport and Highways Nitin Gadkari said on Monday. Speaking at the International Workshop on 'Sustainable Tunnelling for Better Life' held at MIT World Peace University (MIT-WPU), Gadkari called for a renewed focus on reducing construction costs, upgrading technology, and indigenising tunnelling machinery.

The minister's remarks came during the inauguration of India's first center of excellence for tunnelling and underground construction at MIT-WPU. Developed in partnership with Tata Projects and Sandvik, the centre features a Tunnel Monitoring Laboratory and a Drilling & Blasting Laboratory aimed at bolstering domestic capabilities in underground infrastructure.

"India is entering a golden era of infrastructure. Tunnels are essential for connectivity, safety, and sustainability," Gadkari said. "We must adopt fuels like ethanol, CNG, hydrogen, and electricity, and manufacture or import used tunnel boring machines from Europe to cut costs." The two-day workshop, co-organised with the International Tunnelling and Underground Space Association's Committee on Education and Training (ITA-CET), spotlighted the importance of sustainable practices and advanced engineering in tunnelling - especially in sensitive regions like the Himalayas.

Experts from India, Europe, the UK, and the US deliberated on digital innovations such as Building Information Modeling (BIM), laser scanning, and low-impact excavation techniques. Emphasis was laid on reducing the environmental footprint of tunnels and increasing the safety of underground construction workers.

Arnold Dix, Past President, International Tunnelling Association, underlined the global relevance of India's initiative. "This Centre bridges a critical skills gap. Too often, young workers are at risk because they lack practical training. What I see here at MIT-WPU - the mentorship and family-like academic environment - is exceptional," he said.

India's tunnel projects - ranging from highway expansions to strategic defence and urban mobility - are being fast-tracked to support economic growth, energy security, and disaster resilience. The government is pushing for deeper involvement from private players, global technology providers, and academic institutions to reduce imports and enhance local skill development.

Gadkari assured that his ministry would back initiatives with equipment and training support. "With innovation, commitment, and collaboration, we can make India self-reliant in tunnelling technology," he said.

The Indian EXPRESS
JOURNALISM OF COURAGE

News / Cities / Pune / Tunnel projects worth Rs 3L crore in next 10 yrs in India: Nitin Gadkari

Tunnel projects worth Rs 3L crore in next 10 yrs in India: Nitin Gadkari

A Center of Excellence for Tunnelling and Underground Construction was also inaugurated at MIT-WPU, featuring a Tunnel Monitoring Laboratory and a Drilling & Blasting Laboratory

Written by Soham Shah
Pune | June 24, 2025 00:01 IST

NewsGuard

Union Minister of Road Transport and Highways Nitin Gadkari said that tunnelling projects worth Rs 2.5 to Rs 3 lakh crore would be built in India over the next 10 years. Speaking at the inauguration of the International Workshop on Sustainable Tunnelling for Better Life at MIT World Peace University in Pune Gadkari said, "There is big potential in tunnels. Over the next 10 years, we plan to build tunnel projects worth Rs 2.5 to Rs 3 lakh crore. In Delhi under the Talkatora stadium to Gurgaon we have started the construction. In Bangalore we have started making a tunnel from both the sides of the city."

He added, "Many tunnelling machines come from China. Bringing spare parts and machines from China is a difficulty. I don't want to go into the details of that. But we will have to think about how to manufacture machines in our own country… To make this a reality, we must reduce construction costs without compromising quality. That means using new technologies and sustainable fuels like CNG, ethanol, hydrogen, and electric alternatives to diesel. Using diesel in drilling machines has to stop. We should also refurbish old tunnelling machines, import used ones from European countries like Austria, Norway, and Spain, and eventually manufacture our own."

A Center of Excellence for Tunnelling and Underground Construction was also inaugurated at MIT-WPU, featuring a Tunnel Monitoring Laboratory and a Drilling & Blasting Laboratory. The Center of Excellence, set up in collaboration with Sandvik and Tata Projects Ltd., aims to support advanced research and training in underground construction technologies.

Arnold Dix, former president of the International Tunnelling Association, highlighted the importance of having practical experience before graduates are sent out into the workforce of tunnelling. " For decades I have attended disasters and I know young men and women who lose their lives because they just don't know how to build the tunnels. And they work so hard but the skills section is to be absolutely supported, it is a wonderful thing. Too often, young workers are placed at risk because they lack the training needed to safely construct what has been so carefully designed."

S K Dharmadhikari, an infrastructure sector veteran, was conferred with the Lifetime Achievement Award for his invaluable contribution in nurturing and shaping the next generation of tunnel engineers. Companies like TATA Projects Ltd, Larsen & Toubro, Nagpur Mumbai Super Communication Expressway Limited (NMSCEL), J. Kumar Infra projects Ltd were awarded with different categories of awards.


The Indian EXPRESS
JOURNALISM OF COURAGE

MIT-WPU develops tech to produce hydrogen, BioCNG from agri waste

The research demonstrates success with mixed agro waste, including millet trash and other seasonal crop residue. Express

EXPRESS NEWS SERVICE
PUNE, SEPTEMBER 12

supported by four granted patents, is now established on the MIT-WPU campus. The generated biogas showed high

PRESS TRUST OF INDIA
India's premier news agency

Researchers claim carbon-negative tech development for green hydrogen production

MUMBAI: Sept 10 Researchers at MIT World Peace University (MIT-WPU), Pune, claimed to have developed a carbon-negative process that produces both bioCNG and green hydrogen from mixed agricultural waste.

This research offers a cleaner and more affordable path to energy independence, according to the Green Hydrogen Research Centre at MIT-WPU.

This process aligns with the National Green Hydrogen Mission, which aims to produce 5 million tonnes of Green Hydrogen annually by 2030, MIT-WPU said in a statement on Wednesday.

Pioneering sustainable innovation—MIT-WPU researchers develop carbon-negative technology to produce green hydrogen and BioCNG from agricultural waste, gaining national media recognition.

Hindustan Times 100

MIT-WPU signs MoU with VLSI society to boost semiconductor startups

By [illegible]

Published on: Jul 31, 2025 10:04 am IST

India's semiconductor sector is poised to generate upto 10 lakh new jobs over the next five years, signalling a major stride towards economic self-reliance and technological advancement, according to the president of VLSI Society of India, Satya Gupta.

MIT-WPU signs MoU with VLSI society to boost semiconductor startups

Pune: India's semiconductor sector is poised to generate upto 10 lakh new jobs over the next five years, signalling a major stride towards economic self-reliance and technological advancement, according to the president of VLSI Society of India, Satya Gupta. He made this remark on July 29 while addressing a gathering of 250 delegates at the Semiconductor Conclave 2025, held at MIT World Peace University in Pune.

"From high-end chip design to technician and manufacturing jobs, this growth will empower our youth and help position India as a global electronics and semiconductor hub by 2047," said Gupta to over 250 delegates from academia, government, and industry.

The conclave, organised by MIT-WPU's Department of Electrical and Electronics Engineering (DoEEE) with support from the Department of Scientific and Industrial Research (DSIR), and Ministry of Science and Technology provided a vital platform for discussion on India's growing semiconductor ecosystem.

A Memorandum of Understanding (MoU) was signed between MIT-WPU and the VLSI Society of India on the occasion to promote research, innovation, and startup development. "This collaboration will strengthen the semiconductor startup ecosystem by providing techno-commercial mentorship, industry linkages, and seed funding support," said Milind Pande, pro vice-chancellor, MIT-WPU.

Strengthening industry-academia collaboration—MIT-WPU signs MoU with VLSI Society of India to accelerate semiconductor innovation and empower future-ready startups.

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